SPANS-04: Service Dependencies & Flow Analysis

> **Series:** SPANS — Distributed Tracing and Spans | **Notebook:** 4 of 8 | **Created:** December 2025 | **Last Updated:** 10/28/2026

Mapping Your Distributed System

This notebook teaches you how to use span data to understand service relationships, analyze request flows, and identify critical dependencies in your system.

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Prerequisites

Before starting this notebook, ensure you have:

- ✅ Completed **SPANS-01** through **SPANS-03**

- ✅ Access to a Dynatrace environment with distributed trace data
- ✅ Understanding of span kinds (server, client, producer, consumer)

SaaS 1.337: OpenTelemetry `service.name` in SDv1 Service Names

Per the SaaS 1.337 release notes, *"You can now use the OTel service.name attribute to customize the name of SDv1 services."* Smartscape shows the result as `service.name (detected name)`, and the change *"affects the service name in all Smartscape use cases, and the dt.service.name attribute on spans and metrics."*

So `dt.service.name` — present on every span, OneAgent and OpenTelemetry alike — carries the enriched name where an OTel `service.name` exists, and the detected name otherwise. That makes it the service field this series groups by:

```
fetch spans, from:-1h
| filter isNotNull(dt.service.name)
| summarize span_count = count(), by:{dt.service.name, dt.smartscape.service}
| sort span_count desc
| limit 50
```

> **Sources:** SaaS 1.337 release notes (DT docs). **Dictionary:** `dt.service.name ( stable ;`
>
> described as equal to the Smartscape service node name), read 10/02/2026.

SaaS 1.344: Frontend nodes in the service dependency graph

SaaS 1.344+ (staged tenant rollout from 07/29/2026). Per the release notes, *"the Service dependency graph now shows FRONTEND nodes"*, and spans with frontend links can be followed into the related user events and sessions from Distributed Tracing.

Two topology-facing changes:

1. **The Smartscape service dependency graph now displays `FRONTEND` nodes.** The frontend that originates a request appears as a node in its own right rather than as an implicit off-graph caller — the topology view starts where the user does, instead of at the first server span it can see.
2. **Distributed tracing can explore frontend events and sessions without switching apps.** A slow backend dependency and the user session that hit it can be read on one surface.

Neither change alters any DQL in this notebook. The `span.kind == "client"` / `server.address` techniques in **Section 3** remain how you *derive* dependencies from span data — and on a pre-1.344 tenant they remain the only way to do it. The graph is a reading surface layered over the same relationships; it does not replace the queries, and a dependency you need in a dashboard tile, a report, or an SLO still has to be queried.

The frontend-to-backend link in span data

The linkage the graph draws on is modelled in the semantic dictionary as **`frontend_link`** and exposed on spans through the **`frontend.link`** record field:

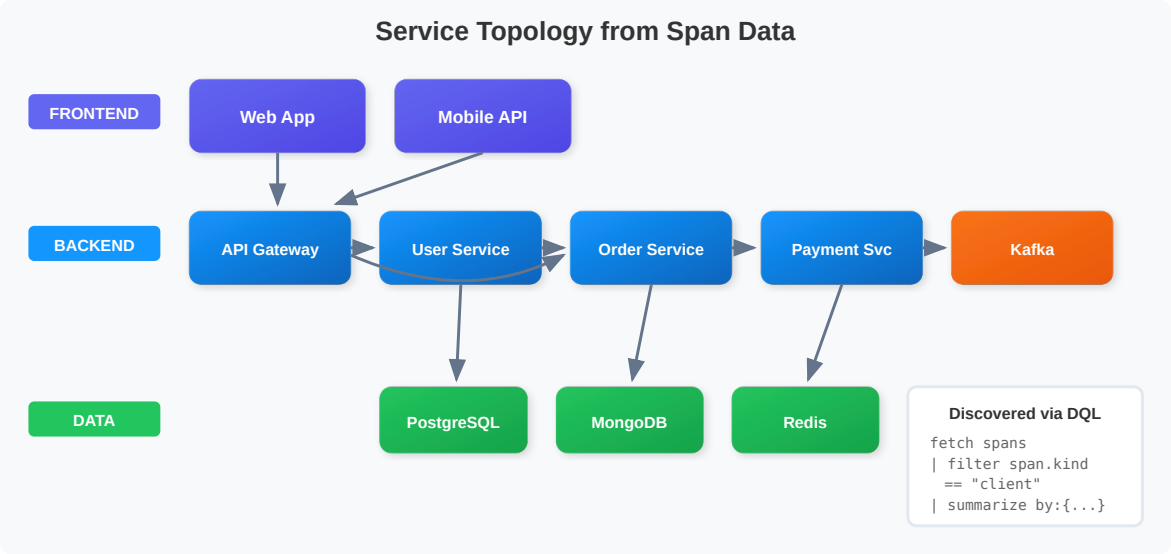
Field	Carries
`span.id`	The backend span the frontend action links to
`dt.rum.session.id`	The RUM session that originated the request
`dt.rum.instance.id`	The RUM instance (page / view load) within that session
`dt.rum.is_linking_candidate`	Whether the span is eligible for frontend linking

Session-side configuration of this linkage belongs to the **WEBRUM** series. For how the session identifier travels between browser and backend, see **OTEL-04 § 4 — Context Propagation**.

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1. Understanding Service Topology

Service topology shows how your services connect and communicate:



Key Span Types for Topology

Span Kind	Role	What It Tells You
`server`	Receives requests	Entry points, inbound traffic
`client`	Makes requests	Outbound calls, dependencies
`producer`	Sends messages	Async event sources
`consumer`	Receives messages	Async event handlers

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2. Discovering Services

First, discover all services generating spans in your environment:

```
// List all services with span counts
fetch spans, from:-1h
| summarize {
    span_count = count(),
    operations = countDistinct(span.name)
  }, by: {dt.service.name}
| sort span_count desc
| limit 30
```

```
// Service role analysis - understand each service's function
fetch spans, from:-1h
| summarize {
    server_spans = countIf(span.kind == "server"),
    client_spans = countIf(span.kind == "client"),
    internal_spans = countIf(span.kind == "internal"),
    producer_spans = countIf(span.kind == "producer"),
    consumer_spans = countIf(span.kind == "consumer")
  }, by: {dt.service.name}
| sort server_spans desc
| limit 30
```

```
// Discover all operations/endpoints per service
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    request_count = count(),
    avg_duration_ms = avg(duration) / 1ms
  }, by: {dt.service.name, span.name}
| sort dt.service.name asc, request_count desc
| limit 50
```

---

3. Mapping Service Dependencies

Use CLIENT spans to understand which services call which other services:

Key Attributes for Dependencies

Attribute	Description
`server.address`	Target host/address (`stable`)
`server.port`	Target port (`stable`)

`peer.service`, which older material uses for the target service, has no row in the semantic dictionary and was set on 0 of 8,286,060 spans over 24 hours on the validation tenant. Map dependencies from `server.address` on client spans, or by joining each client span to the server span it called (same `trace.id`, server `span.parent_id` = client `span.id`). The join below groups by service **name** for readability; service names can

repeat across applications (`frontend` was 4 services on the validation tenant), so add `dt.smartscape.service` on both sides when one name covers several services.

> **Dictionary:** `server.address` (`stable`), `server.port` (`stable`); no row for `peer.service` or `http.host` , read 10/02/2026 (control: `server.address` returned a row).

```
// Map service-to-service calls using CLIENT spans
fetch spans, from:-1h
| filter span.kind == "client"
| summarize {
    call_count = count(),
    avg_latency_ms = avg(duration) / 1ms,
    error_count = countIf(span.status_code == "error")
  }, by: {dt.service.name, span.name}
| fieldsAdd error_rate_pct = (error_count * 100.0) / call_count
| sort call_count desc
| limit 50
```

```
// Which service calls which: join each client span to the server span it
called.
// The server span's parent is the client span (same trace.id).
fetch spans, from:-1h
| filter span.kind == "server" and isNotNull(span.parent_id)
| fields trace.id, callee = dt.service.name, parent = span.parent_id
| join [
    fetch spans, from:-1h
    | filter span.kind == "client"
    | fields trace.id, caller = dt.service.name, client_span = span.id,
call_duration = duration
  ], on:{trace.id, left[parent] == right[client_span]}, fields:{caller,
call_duration}
| filter caller != callee
| summarize {call_count = count(), avg_latency_ms = avg(call_duration) / 1ms},
by:{caller, callee}
| sort call_count desc
| limit 30
```

```
// Map dependencies using HTTP host/URL information
fetch spans, from:-1h
| filter span.kind == "client" and isNotNull(server.address)
| summarize {
    call_count = count(),
    avg_latency_ms = avg(duration) / 1ms,
    error_count = countIf(span.status_code == "error")
  }, by: {dt.service.name, server.address}
| fieldsAdd error_rate_pct = (error_count * 100.0) / call_count
| sort call_count desc
| limit 30
```

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4. Client-Server Call Patterns

Analyze the matching CLIENT and SERVER span pairs to understand inter-service communication:

```
// Analyze outbound calls from each service
fetch spans, from:-1h
| filter span.kind == "client"
| summarize {
    outbound_calls = count(),
    avg_latency_ms = avg(duration) / 1ms,
    p99_latency_ms = percentile(duration, 99) / 1ms,
    error_count = countIf(span.status_code == "error")
  }, by: {dt.service.name}
| fieldsAdd error_rate_pct = (error_count * 100.0) / outbound_calls
| sort outbound_calls desc
| limit 20
```

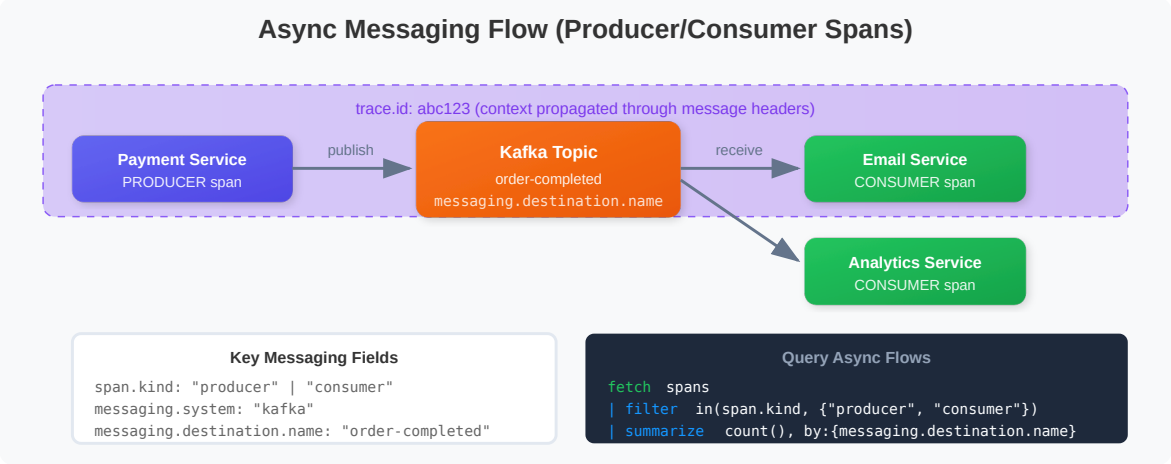
```
// Analyze inbound requests to each service
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    inbound_requests = count(),
    avg_latency_ms = avg(duration) / 1ms,
    p99_latency_ms = percentile(duration, 99) / 1ms,
    error_count = countIf(span.status_code == "error" or
http.response.status_code &gt;= 500)
  }, by: {dt.service.name}
| fieldsAdd error_rate_pct = (error_count * 100.0) / inbound_requests
| sort inbound_requests desc
| limit 20
```

```
// Find services with high outbound/inbound ratio (integration heavy)
fetch spans, from:-1h
| summarize {
    inbound = countIf(span.kind == "server"),
    outbound = countIf(span.kind == "client")
  }, by: {dt.service.name}
| filter inbound &gt; 0
| fieldsAdd outbound_ratio = toDouble(outbound) / toDouble(inbound)
| sort outbound_ratio desc
| limit 20
```

---

5. Async Messaging Flows

Analyze asynchronous messaging patterns using PRODUCER and CONSUMER spans:



Key Messaging Attributes

Attribute	Description
`messaging.system`	Kafka, RabbitMQ, etc.
`messaging.destination.name`	Topic/queue name
`messaging.operation.type`	publish, receive, etc. (`messaging.operation` has no dictionary row)

Trace Context Across Message Brokers

PRODUCER and CONSUMER spans only join the **same trace** if trace context travels inside the message. Whether that happens automatically depends on the client library:

Broker	Auto-instrumentation	Implication
Kafka, RabbitMQ	First-class — OpenTelemetry Java lists Kafka `0.11+` and RabbitMQ `2.7+`	`traceparent` is injected and extracted for you
Google Cloud Pub/Sub	**Absent from that supported-libraries list**	Context is **not** propagated for you — the trace breaks silently at each hop

Where auto-instrumentation is missing, carry the W3C `traceparent` yourself through the message `attributes` map:

- **Publisher:** inject the active context as a `traceparent` attribute before publishing.
- **Subscriber:** read `traceparent` from the message attributes and use it as the parent when starting the consumer span.

Symptom to look for: consumer spans that are *root* spans. If the queries below show healthy producer **and** consumer volume for a destination but the two never appear in one trace, missing context propagation is the usual cause.

> Auto-instrumentation coverage varies by language and client version — verify for your own runtime rather than assuming. **Sources:** Span and trace context propagation (DT docs); OpenTelemetry Java supported libraries (OpenTelemetry GitHub)

```
// Find message producers (services sending async messages)
fetch spans, from:-1h
| filter span.kind == "producer"
| summarize {
    messages_sent = count(),
    avg_duration_ms = avg(duration) / 1ms,
    error_count = countIf(span.status_code == "error")
  }, by: {dt.service.name, span.name}
| sort messages_sent desc
| limit 20
```

```
// Find message consumers (services receiving async messages)
fetch spans, from:-1h
| filter span.kind == "consumer"
| summarize {
    messages_received = count(),
    avg_processing_ms = avg(duration) / 1ms,
    error_count = countIf(span.status_code == "error")
  }, by: {dt.service.name, span.name}
| fieldsAdd error_rate_pct = (error_count * 100.0) / messages_received
| sort messages_received desc
| limit 20
```

```
// Analyze messaging system usage (Kafka, RabbitMQ, etc.)
fetch spans, from:-1h
| filter span.kind == "producer" or span.kind == "consumer"
| filter isNotNull(messaging.system)
| summarize {
    message_count = count(),
    producers = countIf(span.kind == "producer"),
    consumers = countIf(span.kind == "consumer"),
    error_count = countIf(span.status_code == "error")
  }, by: {messaging.system, messaging.destination.name}
| sort message_count desc
| limit 20
```

```
// Map producer to consumer relationships
fetch spans, from:-1h
| filter span.kind == "producer" or span.kind == "consumer"
| summarize {
    span_count = count()
  }, by: {dt.service.name, span.kind, messaging.destination.name}
| sort messaging.destination.name, span.kind
| limit 30
```

6. Trace Hierarchy Analysis

Analyze parent-child relationships within traces to understand call depth:

```
// Count spans per trace to understand trace complexity
// trace_duration_ms is wall-clock (last end minus first start), not a sum of
span durations.
fetch spans, from:-1h
| summarize {
    span_count = count(),
    services_involved = countDistinct(dt.service.name),
    first_start = min(start_time),
    last_end = max(end_time)
  }, by: {trace.id}
| fieldsAdd trace_duration_ms = (last_end - first_start) / 1ms
| fields trace.id, span_count, services_involved, trace_duration_ms
| sort span_count desc
| limit 25
```

```
// Examine a complete trace hierarchy
// Replace YOUR_TRACE_ID with an actual trace ID from above
fetch spans, from:-1h
// trace.id is a uid: keep the toUid() wrapper — a plain string never matches.
// | filter trace.id == toUid("YOUR_TRACE_ID")
| fieldsAdd duration_ms = duration / 1ms
| fields start_time,
         span.id,
         span.parent_id,
         dt.service.name,
         span.name,
         span.kind,
         duration_ms
| sort start_time asc
| limit 100
```

```
// Find entry points (root spans) and their downstream services
fetch spans, from:-1h
| filter isNull(span.parent_id)
| summarize {
    entry_count = count(),
    avg_duration_ms = avg(duration) / 1ms
  }, by: {dt.service.name, span.name}
| sort entry_count desc
| limit 20
```

---

7. Cross-Service Latency Analysis

Identify latency hot spots between services:

```
// Latency by service-to-service call
fetch spans, from:-1h
| filter span.kind == "client" and isNotNull(server.address)
| summarize {
    call_count = count(),
    avg_ms = avg(duration) / 1ms,
    p95_ms = percentile(duration, 95) / 1ms,
    p99_ms = percentile(duration, 99) / 1ms
  }, by: {dt.service.name, server.address}
| sort p95_ms desc
| limit 20
```

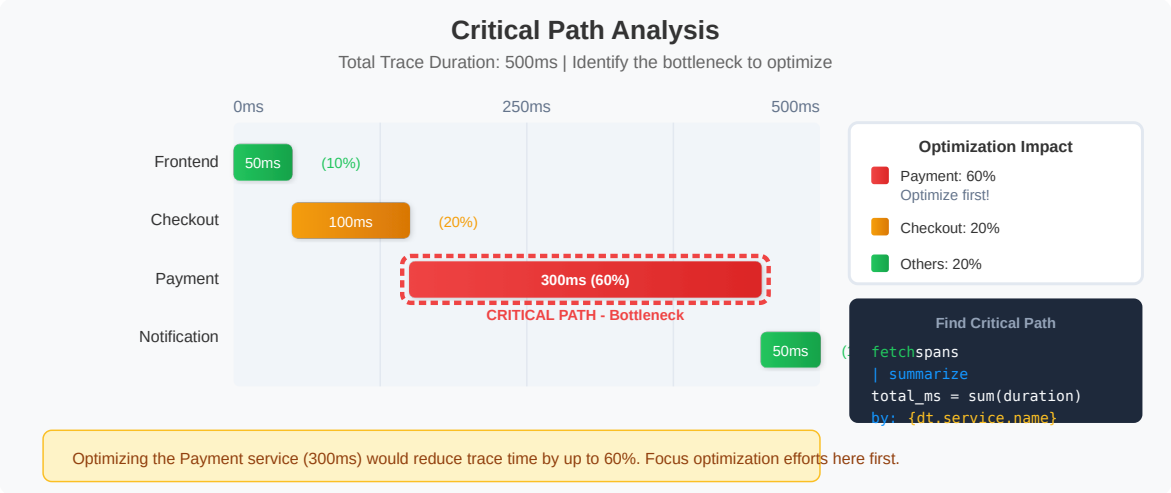
```
// Time spent per service in traces
fetch spans, from:-1h
| summarize {
    total_time_ms = sum(duration) / 1ms,
    span_count = count(),
    avg_per_span_ms = avg(duration) / 1ms
  }, by: {dt.service.name}
| sort total_time_ms desc
| limit 20
```

```
// Find slowest dependencies (CLIENT spans)
fetch spans, from:-1h
| filter span.kind == "client"
| filter duration &gt; 500ms
| summarize {
    slow_call_count = count(),
    avg_duration_ms = avg(duration) / 1ms,
    max_duration_ms = max(duration) / 1ms
  }, by: {dt.service.name, span.name}
| sort avg_duration_ms desc
| limit 20
```

---

8. Critical Path Analysis

Identify the services and operations that contribute most to end-to-end latency:



```
// Find services contributing most span time.
// sum(duration) is INCLUSIVE time: a span's duration includes the child calls
it waits on,
// so a caller is credited with its callees' time as well. Use it to rank
where time is
// spent, not as each service's own (self) time.
fetch spans, from:-1h
| summarize {
    total_span_time_ms = sum(duration) / 1ms,
    span_count = count(),
    avg_duration_ms = avg(duration) / 1ms,
    max_duration_ms = max(duration) / 1ms
  }, by: {dt.service.name}
| sort total_span_time_ms desc
| limit 15
```

```
// Find slowest operations across all services
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    call_count = count(),
    avg_duration_ms = avg(duration) / 1ms,
    p99_duration_ms = percentile(duration, 99) / 1ms,
    total_time_ms = sum(duration) / 1ms
  }, by: {dt.service.name, span.name}
| filter call_count &gt; 10
| sort p99_duration_ms desc
| limit 20
```

```
// Identify high-impact optimization candidates
// (high volume + high latency = most benefit from optimization)
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    call_count = count(),
    avg_duration_ms = avg(duration) / 1ms,
    total_time_ms = sum(duration) / 1ms
  }, by: {dt.service.name, span.name}
| filter call_count &gt; 50
| fieldsAdd impact_score = call_count * avg_duration_ms
| sort impact_score desc
| limit 20
```

Summary

In this notebook, you learned:

✅ **Service discovery** - Find all services and their operations
✅ **Dependency mapping** - Use CLIENT spans with `server.address`, or join client spans to the server spans they called
✅ **Client-server patterns** - Analyze inbound/outbound call ratios
✅ **Async messaging** - Track PRODUCER/CONSUMER spans through message queues
✅ **Trace hierarchy** - Understand span parent-child relationships
✅ **Cross-service latency** - Find latency hot spots between services
✅ **Critical path analysis** - Identify bottlenecks for optimization

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Next Steps

Continue to **SPANS-05: Advanced Span Analytics** to learn:
- Time series analysis and trending
- Complex aggregations and calculations
- Building dashboard-ready queries
- Alerting patterns

References

- Trace semantic conventions (DT docs)
- SaaS 1.337 release notes (DT docs)
- SaaS 1.344 release notes (DT docs)
- Span and trace context propagation (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.